

Collatz-Conjecture Proved and Halemane-Conjecture Proposed

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The entire dynamics of the Collatz $(3n+1)$ System can be exactly represented by a dynamically-evolving graded-algebraic-structure of an ideal-based-filtration-scheme; with divisibility-classes for modulo-3 quotient-semiring generated by the primitive root 2; along with a filtration-shifting-global-affine-transformation, $f(x)=(3x+1)$ when x is a positive odd number; achieving a Euclidean-expansion-shift to the topmost coprime-layer of that filtration-scheme; while also avoiding the modulo-multiple-layer (zero-layer) and all the nilpotent-layers with nilpotent-elements (dead-end zero-divisors) like $(6m-3)$ and $(6m-0)$; the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ being bypassed by an initialization-phase for this filtration scheme, starting directly with the modul-9 coprime-layer. The transitive-closure of the Collatz $(3n+1)$ System is exactly represented by the transitive-closure of this dynamically evolving algebraic system which is indeed the complete set of natural numbers. This design for the system-dynamics-framework by itself provides a definitive-proof of the Collatz $(3n+1)$ Conjecture, establishing that the entire Collatz Map is exactly represented by this dynamically-evolving graded-algebraic-structure defined on the set of all natural numbers; neither missing any natural-number nor including any extraneous-elements; providing both the necessary-&-sufficient condition simultaneously in one single sweep.

Halemane-Conjecture states that the maximum number of odd $(3n+1)$ operations required to reach the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ starting from any given positive integer and moving along the Collatz sequence, is limited by that given number itself; with the triad $\{(31 \Leftarrow 41 \Leftarrow 27)\}$ as an exceptional limiting case.